

Aditya Khamitkar

AI/ML Systems Engineer

LLM Evaluation | Agentic AI | Generative AI | MLOps | Scalable Data Systems

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Summary

AI/ML Systems Engineer (Integrated M.Tech, June 2027) with about 2 years of hands-on experience (Oct 2024 to present) in LLM evaluation and benchmarking, agentic AI and ML data systems: frontier-lab evaluation work at Deccan AI, production-level AI agents at Bosch, and telemetry modelling at ISRO. Scripted Python audits against live SQLite databases to resolve schema bugs across an 18-tool agentic RL environment and delivered **300+ hours** of human-in-the-loop RLHF and SFT evaluation. Author of a survey manuscript on model collapse from recursive training on synthetic data, with a further manuscript in preparation on BDH ablations. Also building Atlas, an open-source C++20 compute engine (in development). Strong foundations in distributed and scalable systems for data science and ML, big data analytics and data warehousing.

Skills

Core Areas: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Agentic AI, LLM evaluation and benchmarking, distributed and scalable systems for data science and ML, machine learning for large datasets, optimisation methods for ML, big data analytics, advanced data analytics, data mining, data warehousing

Languages: Python, C++ (17/20), SQL, Bash

LLM & Agentic AI: LLM evaluation, RLHF (rubric-based grading, side-by-side comparison), prompt engineering, RAG pipelines, multi-agent systems, LangGraph, LangChain, Pydantic structured outputs, Hugging Face Transformers, Groq API, embeddings and outlier analysis

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LSTM, GANs, state-space models (SSMs), time-series forecasting, Bayesian optimisation, cross-validation

Data & Big Data: Pandas, NumPy, SciPy, Apache Spark, PySpark, Spark SQL, ETL pipelines, data warehousing, data mining, EDA, hypothesis testing, A/B testing, Tableau, Power BI, Matplotlib, Seaborn, Streamlit

Cloud & MLOps: AWS (EC2, S3 hands-on; wider services via coursework), Microsoft Azure (AZ-900 fundamentals level), Docker, Kubernetes, MLflow, FastAPI, REST APIs, Git, Linux, SQLite

Systems & Engineering: C++ RAII and memory management, Pybind11, multithreading fundamentals, data structures and algorithms, OOP, SOLID, design patterns, clean architecture, SDLC, Agile

Work Experience

Deccan AI Experts

AI Training Specialist, LLM Evaluation & Agentic Environments (Freelance)

Jan 2025 - Present

Remote

Project-based work on frontier-lab post-training and agentic RL-environment projects

- Evaluated SQL and tool-using agents on an **18-tool agentic RL environment**: scripted automated Python audits against live SQLite databases to find and resolve schema bugs and structural anomalies before deployment.
- Delivered **300+ hours** of human-in-the-loop RLHF and SFT evaluation, scoring outputs against rubrics for factual accuracy, verbosity and instruction adherence, with side-by-side (SxS) comparisons and clear written feedback, in an Agile, sprint-based delivery environment.
- Evaluated multi-agent workflows and flagged alignment failures before deployment.
- Ran systematic EDA, distribution analysis and outlier detection (Pandas, NumPy) across **3 production RLHF corpora** to detect and flag data-integrity violations before fine-tuning, supporting data-governance standards.
- Collaborated with AI researchers and engineering teams on model evaluation, dataset curation and schema verification for production AI systems.

Bosch Global Software Technologies (BGSW)

Software Development Engineer Intern (Student Trainee), Enterprise Applications for Multi Market

Aug 2026 - May 2027

Pune, India

- Working on AI and data for Bosch Enterprise Applications for Multi Market: building production-level, scalable AI agents as part of a 10-month Software Development Engineering internship.

Outlier, Turing, Bespoke Labs

Freelance LLM Evaluation & Benchmarking (project-based)

2025 - 2026

Remote

- Delivered project-based LLM evaluation and benchmarking work on a per-assignment basis.

- Cut resume-screening time by **80%** (50 resumes in 42 s vs. 120 min) by building an end-to-end AI parsing and scoring platform with SpaCy NLP pipelines, pdfplumber and Streamlit; reached **90%+** candidate-categorisation accuracy with custom NLP scoring.

- Reduced manual signal-review effort by **40%** by building PyTorch and TensorFlow LSTM time-series models across **4 PSLV/SSLV launch-vehicle telemetry streams** (RCS, yaw, pitch, roll) to quantify signal degradation for launch-safety analysis.
- Engineered multi-source features with stationarity profiling and missing-data imputation to counter real-time atmospheric interference in a radar signal-strength prediction system.
- Authored 4 technical reports (Matplotlib, Seaborn) documenting model methodology and pipeline performance for cross-functional engineering teams.

Projects

Atlas: Distributed Compute Engine for ML & Data Workloads

2026 - Present

C++20, Python, Pybind11 | github.com/TechieSamosa/Atlas

- In active development: an open-source C++20 execution engine with a Python SDK for ML and data-intensive workloads (thread pool, custom allocators, task scheduler and Python SDK are on the published roadmap).
- Wrote **The Atlas Codex, Volume I**: a 100+ page engineering manual on C++20 memory models and RAII, custom allocators, concurrency and the C++/Python boundary, with hands-on tutorials (arena allocator, pool allocator, thread pool with future-based results).

Nexus Scout: Multi-Agent Talent Acquisition Pipeline

LangGraph, LangChain, Groq API, Pydantic, Streamlit | github.com/TechieSamosa/nexus-scout-catalyst | [live demo](#)

- Designed a deterministic multi-agent LangGraph orchestration with agent routing and structured Pydantic output validation; a weighted hybrid scoring engine (**60%** technical fit, **40%** interest alignment) automates personalised outreach at scale.

Project AETHER: Lunar PSR Imaging & AI Pipeline

2024 - Present

PyTorch, GDAL, Rasterio, GANs, HuggingFace Accelerate | github.com/TechieSamosa/Aether

- Improved SNR in lunar Permanently Shadowed Regions by **+86 dB** using a custom GAN fine-tuned on a self-supervised masked-autoencoder (MAE) pre-training stage, enabling visibility in zero-illumination craters.
- Built a GDAL/Rasterio ETL and data-fusion pipeline over multi-instrument Chandrayaan-2 data (OHRC, DFSAR, CLASS, IIRS; SPICE kernels), curating **38,000+** georeferenced patches into multi-channel training tensors.

Prakshep Optimization Engine: Aerospace Digital Twin & Flight Simulation

2026 - Present

C++17, WebGPU, FastAPI, PyTorch

- Engineered a custom 6-DOF flight-physics core in C++ for a real-time flight-simulation and digital-twin platform for orbital launch vehicles, with WebGPU rendering and streaming telemetry.

Synapse.cpp: Neural-Network Engine from Scratch

C++17, RAII, Adam/SGD, zero external dependencies | github.com/TechieSamosa/Synapse.cpp

- Implemented forward/backward propagation, Adam with bias correction, SGD, dropout and gradient clipping in raw C++17; RAII-compliant Matrix class with move semantics and row-major flat storage, validated across **4 benchmarks**.

AgriSat AI: Distributed Crop-Health Analytics

2024

PySpark, S3-compatible object storage, scikit-learn, Tableau | [live demo](#)

- Attained **92%+** classification accuracy and **35%** runtime reduction with a soft-voting ensemble (Random Forest, Gradient Boosting, SVM) on NDVI/EVI spectral features via a distributed PySpark ETL pipeline.

DentalIOS: Offline-First Dental Practice Management Platform

2026 - Present

Python, PySide6, SQLite

- Developing a modular, offline-first desktop application for clinical management with a maintainable, layered architecture.

Publications & Research

Alception: Recursive Training of Generative AI on Synthetic Data

A. Khamitkar (sole author) | Survey manuscript in preparation

2026

- Technical survey of model collapse under recursive training, building on Shumailov et al. (Nature, 2024) and Gerstgrasser et al. (COLM, 2024): early tail erosion vs late homogenisation, data-replacement vs data-accumulation workflows, and mitigations (provenance tracking, watermarking, AI-content detection, contamination-free dynamic benchmarking).
- Proposes a controlled experimental framework (decoder-only transformer, three training conditions, diversity, distributional-fidelity and tail-coverage metrics) for future empirical study.

BDH Architecture Analysis: A Controlled Component-Level Study of the Dragon Hatchling Language Model

A. Khamitkar, T. Jagatap, N. Saini | Manuscript in preparation | PyTorch | [code](#)

2026

- Controlled 5-variant ablation of Pathway's Dragon Hatchling (BDH) architecture (Kosowski et al., arXiv:2509.26507) on byte-level WikiText-2, including a parameter-matched Transformer baseline.
- Removing the multiplicative gate raised loss by **0.059 nats** (largest effect of the components tested); 4x latent compression (m: 128 to 32) lowered loss by **0.052 nats**; ReLU to GELU lowered it by 0.039 nats.

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Education

VIT Bhopal University

Sep 2022 - Jun 2027 (Expected)

Integrated M.Tech (5-year Master's programme), Computer Science & Engineering (Computational & Data Science)

- Relevant coursework: Distributed and Scalable Systems for Data Science, Machine Learning for Large Datasets, Optimisation Methods for Machine Learning, Big Data Analytics, Advanced Data Analytics, Data Mining and Data Warehousing, Data Engineering, Deep Learning for Computer Vision, Reinforcement Learning, High-Performance Computing, Distributed Systems, Applied Statistical Methods, Healthcare Data Analytics, Cloud Computing (AWS), Database Management Systems, Operating Systems, Computer Networks

Space Central School (ISRO Campus), Sriharikota

CBSE Class XII (2021): 94.4% | CBSE Class X (2019): 89.8%

2019 - 2021

Awards

- **Amazon ML Challenge:** Top 100 Finalist.
- **Intel AI4Youth National Finalist (2021):** top 0.24% (Top 125) of 52,000+ participants with Doctor Bot, a transformer-based healthcare NLP chatbot endorsed by NITI Aayog for deployment in Uttar Pradesh.
- **Smart India Hackathon 2024 Finalist:** Social Media Fraud Prevention track.
- **National Aerolympics 2020:** All-India Champion.

Certifications

- IBM AI Engineering Specialization, IBM / Coursera (2023)
- Machine Learning with Python (with Honors), IBM / Coursera (2023)
- Supervised Machine Learning: Regression & Classification, Stanford / DeepLearning.AI (2023)
- AI/ML for Geodata Analysis, IIRS-ISRO (2024)

Leadership & Community

- **Technical Team Head, Data Science Club, VIT Bhopal University:** ML/AI workshops, peer mentoring and open-source contributions.
- **Active Member, ML Team & AI Club, VIT Bhopal University:** research discussions on LLMs, model interpretability and scalable ML systems.
- **Open-source contributor:** GirlScript Summer of Code, Omdena.

Languages

English (fluent); Marathi (mother tongue); Hindi and Telugu (native); German (basic, currently learning)